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Effects of glyphosate on immune responses and haemocyte DNA damage of Chinese mitten crab, *Eriocheir sinensis*

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ABSTRACT

As a broad-spectrum organophosphorus herbicide, glyphosate is widely utilized around the world. The toxic effects of glyphosate on Chinese mitten crab, *Eriocheir sinensis*, were assessed using immunotoxicity and genotoxicity biomarkers in this study. The results showed that 24 h and 96 h LC₅₀ values of glyphosate for *E. sinensis* were estimated as 461.54 and 97.89 mg/L, respectively, and the safe concentration was 4.4 mg/L. According to the results above, glyphosate was applied at concentrations of 0, 4.4, 9.8, 44 and 98 mg/L, for 96 h in the exposure experiment. Total haemocyte count (THC) and percentage of granulocytes decreased significantly following 6 h exposure to each concentration of glyphosate and tended to gradually stabilize after 12 h except in 4.4 mg/L, which rapidly recovered to a normal level in 12 h. Phagocytic activity in all treatments decreased dramatically at 6 h and maintained stability until the 96-h mark. Comet tail has been observed early at 24 h in each treatment, and the comet ratio and percentage of DNA (% DNA) in the tail increased as the exposure experiment progressed. Immune-related enzyme activity varied during the experiment. Acid phosphatase (ACP) and alkaline phosphatase (AKP) activities in 44 and 98 mg/L treatments decreased significantly after 48 h exposure, while AKP activities in all concentrations increased markedly at the beginning of exposure. The superoxide dismutase (SOD) and peroxidase (POD) activities increased significantly after 6 h exposure to 44 and 98 mg/L of glyphosates but decreased at 24 h. In addition, the β -glucuronidase (β -GD) activities in the 9.8, 44 and 98 mg/L groups, increased after 6-h exposure and were significantly higher than those in the control at 96 h. These results indicated that glyphosate has evident toxic effect on *E. sinensis* by immune inhibition that is possibly due to the haemocyte DNA damage and a sharp decline in haemocyte numbers, which subsequently induced changes in activities of immune-related enzymes and haemocyte phagocytosis.

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1. Introduction

As a broad-spectrum organophosphorus herbicide, glyphosate-based herbicides, such as the commercial formulations Roundup and Rodeo, are utilized globally and used in both agricultural areas and aquatic waters for weed control [1]. Glyphosate is considered to have low toxicity to animals due to the absence of the shikimate pathway in non-target organisms, which glyphosate depends on to inhibit the enzyme 5-enolpyruvylshikimate-3-phosphate synthase

(EPSPS) [2]. However, high levels of glyphosate have been thought to possibly alter mitochondrial activity in animals by the uncoupling of oxidative phosphorylation during cellular respiration, and mounting evidence suggests that non-target species could be affected [3]. Because of its low persistence, repeated application becomes necessary for weed control, and the residue of glyphosate in aquatic water could reach up to 6.286 mg/L in one day after spraying in a commercial pond [4]. Thus, there will be a potential threat for aquaculture in an area that uses glyphosate. In recent decades, a number of studies have been done to evaluate the toxicity of glyphosate to aquatic organisms. However, until now, most of the studies focused on amphibians, microcrustaceans and fishes, such as *Xenopus*, *Caridina nilotica* and *Anabas testudineus* [5–7], and there are no reports in economically important

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crustaceans such as crabs and shrimps.

The Chinese mitten crab, *Eriocheir sinensis*, is an indigenous freshwater species widely bred in China. The farming of *E. sinensis* has been developing rapidly over the recent two decades with several aquaculture modes. Rice-crab coculture is an important and widespread mode in many places in China, especially in Liaoning Province [8]. However, the contradiction between glyphosate use and crab breeding in agricultural areas became more obvious. Therefore, the effects of glyphosate on *E. sinensis*, including acute toxicity and immune injury, are necessary to investigate to provide information for the safety assessment of aquaculture. Haemocytes plays an important role in the innate immunity of crustaceans. Previous studies showed that THC, DHC and phagocytosis of haemocytes in the haemolymph, were usually detected as steady immune parameters to evaluate the toxicity effects of exogenous factors, including pathogens, heavy metals, environmental pollutants and pesticides [9–12]. Comet assay was generally employed in assessment of DNA damage from environmental hazard in both terrestrial and aquatic animals [13]. However, little work has been reported about the assessment of toxicity of organophosphorus herbicide to DNA damage in crustaceans using comet assay. Immune-related enzymes such as acid phosphatase, alkaline phosphatase, superoxide dismutase, peroxidase and β -glucuronidase, all participated in immune responses and, except β -glucuronidase, used to be common parameters for toxicity investigation in crustaceans [14–16]. In this study, LC₅₀ values of glyphosate and all parameters above, especially β -glucuronidase, were determined to investigate the immune response under glyphosate stress in *E. sinensis* for the first time, which could provide a foundation for further studies on the establishment of a toxicity assessment system with proper parameters for organophosphorus herbicide in crabs.

2. Materials and methods

2.1. Animals

The experimental crab *Eriocheir sinensis* (Crustacea: Decapoda: Grapsidae) were obtained from a commercial farm in Jintan, Jiangsu Province, China. They were kept in a recirculating system containing filtered freshwater and a UV-treated PVC tube as shelter, with controlled light (12 h in light; 12 h in dark) and regulated temperature (20–22 °C). The animals were fed daily with a commercial crab diet and acclimated in the exposure tank for at least 7 days before the beginning of the experiment. Adult crabs weighing 109 ± 11 g were used for the study.

2.2. Median lethal concentration (LC₅₀) of glyphosate on *E. sinensis*

An acute glyphosate toxicity test was conducted after pre-experiment testing, which progressively narrowed the ranges of concentration between 0% and 100% mortality. Test solutions were prepared by diluting a commercial glyphosate aqueous solution (active ingredient 41% with isopropylamine salt, MeiFeng Chemical Co. Ltd, Jiangsu) to concentrations of 0, 50, 84.14, 141.25, 237.14 and 400 mg/L with distilled water. Each concentration contained 10 crabs and was replicated three times. Test animals were haphazardly selected from all available healthy crabs and randomly allocated to a 125 × 60 × 60 cm plastic aquarium filled with 50 L of fresh water, with 10 crabs per container. A static experimental method was adopted according to the institutional ethical guidelines to determine the 24-h and 48-h median lethal concentrations of glyphosate for *E. sinensis*. No feeding, aeration, or mixing occurred during test time, and water quality parameters, including temperature, pH and dissolved oxygen, were monitored every 24 h.

Dead crabs were classified by no movement of appendages or eyestalks in response to a stick poke. Dead individuals were removed immediately. The LC₅₀ and 95% confidence limits were calculated by a regression probit analysis method in SPSS V20.0 software.

2.3. Glyphosate exposure experiment

According to the 24-h and 48-h LC₅₀ values and the safe concentration of glyphosate on *E. sinensis*, five concentration groups (0, 4.4, 9.8, 44, and 98 mg/L) were set in the exposure experiment. Each group was replicated three times with 10 crabs, with a total of 150 crabs used. Water in the aquarium was aerated constantly and renewed daily by adding fresh water containing the same concentration of glyphosate to ensure that the concentration of glyphosate in each group did not vary during the experiment. Crabs were fed daily and other experimental conditions (light, temperature and water quality) were maintained as mentioned above.

2.4. Haemolymph sampling

Five individuals were randomly taken from each aquarium at 0 h, 6 h, 12 h, 24 h, 48 h and 96 h after the exposure. Haemolymph was drawn with a sterile 1-ml syringe from the unsclerotized membrane of the fifth walking leg. Some haemolymph was diluted at a 1:1 ratio with sterile anticoagulation (trisodium citrate 30 mM, NaCl 338 mM, glucose 115 mM, EDTA 10 mM) immediately to determine the total haemocyte count (THC) and the different haemocyte count (DHC). Some was centrifuged at 1200 r/min for 5 min to collect haemocytes for use in phagocytic activity assay and comet assay. The rest was stored at 4 °C for use in the subsequent assay of immune-related enzyme activity, including acid phosphatase (ACP), alkaline phosphatase (AKP), superoxide dismutase (SOD), peroxidase (POD) and β -glucuronidase (β -GD).

2.5. Total haemocyte count (THC) and different haemocyte count (DHC)

The THC was performed by a haemocytometer under a Motic AE21 inverted phase-contrast microscope (Motic, Canada). Three replicates were made for each sample, and mean value was recorded for statistical analysis.

For DHC detecting, aliquot of 20 μ l haemolymph was dropped onto a clean slide and fixed by a fixative (1% glutaraldehyde and 1% sucrose) for 30 min at 4 °C then dyed with Giemsa solution (Sigma) for 5 min and rinsed in distilled water 3 times. Each slide was air dried and observed under a Motic AE21 inverted microscope with an oil lens (10 × 100). Haemocytes from *E. sinensis* were classified according to prior research [9,17]. Each slide for DHC was examined for at least 5 fields with 200 cells, and the percentages of granulocytes and hyalinocytes were presented as numbers of total haemocytes dividing by numbers of granulocytes or hyalinocytes.

2.6. Phagocytic activity

The phagocytotic activity was investigated using a modified method from Xian et al. [10]. Haemocytes collected by centrifugation at 1200 r/min for 5 min were resuspended in crab saline (NaCl 496 mM, KCl 10 mM, CaCl₂ 16 mM, NaHCO₃ 6 mM, MgSO₄·6H₂O 12 mM, MgCl₂·6H₂O 8 mM, HEPES 20 mM) with a density of 1×10^6 ml⁻¹. Aliquots of 1-ml diluted cell suspensions were mixed with yellow-green fluorescent latex beads (particle size: 1 μ m, density: 4.57×10^8 per milliliter, Sigma), seeded into 24-well culture plates (Corning, USA), and incubated in a dark, moist incubator at 28 °C for 2 h. Haemocytes were collected by a cell scraper,

washed twice with crab saline and centrifuged at 1200 r/min for 3 min to remove non-ingested latex beads. Cells were fixed in a fixative and observed under a DM-i8 inverted fluorescence microscope (Leica, Germany). Five fields were randomly chosen in each well and the phagocytic ratio was presented as a percentage of phagocytized haemocytes out of total haemocytes.

2.7. Comet assay

The alkaline comet assay was adapted from the method described by Singh et al. [18]. First, frosted microscope slides were coated with 100 μ l 0.8% normal melting agarose (NMA) and covered with a coverslip then immediately stored at 4 °C for 10 min to stabilize the first agarose layer. After collection of the haemocytes, 50 μ l of 1% low-melting agarose (LMA) were mixed with 25 μ l of cell suspension, which were adjusted to a density of 1×10^6 ml⁻¹ and was layered over the first agarose layer. The slide was then covered with a clean coverslip and stored at 4 °C for 10 min. After removal of the coverslip, 75 μ l of 0.8% LMA were carefully added onto the layers and cooled at 4 °C for 30 min. Subsequently, slides were placed in a freshly prepared cell lysing solution (2.5 mol/L NaCl, 100 mmol/L Na₂EDTA, 10 mmol/L Tris–HCl, 1% Triton X-100 and 10% DMSO, pH 10) at 4 °C for 40 min. After rinsing in a phosphate buffer solution (PBS) twice, slides were placed in a horizontal electrophoresis chamber covered with alkaline electrophoresis buffer (300 mmol/L NaOH, 1 mmol/L EDTA, pH 13) for 30 min to allow for the unwinding of DNA. Electrophoresis was performed under 25 V and 300 mA for 30 min, and slides were taken out gently and neutralized three times for 5 min in a neutralization buffer (0.4 M Tris–HCl, pH 7.5). Finally, slides were stained with 20 μ l 0.1 mg/ml propidium iodide in a dark, moist chamber for 10 min and observed under a DM-i8 inverted fluorescence microscope. Five fields, with at least 100 cells from each slide, were randomly analysed by the Comet Assay Software Project (CASP), and the DNA damage was expressed as the comet ratio and percentage of DNA in the tail. All steps above were performed under yellow light.

2.8. Immune-related enzyme activity

2.8.1. Serum sampling

Firstly, haemolymph that was pre-cooled in a refrigerator (4 °C) was centrifuged at 1,1000 r/min at 4 °C for 30 min, and the supernatant was carefully collected using a pipette (Eppendorf, Germany). The serum from each crab was then pooled into 2-ml sterile centrifuge tubes and stored at 4 °C for the enzymatic activity assay described below.

2.8.2. Acid phosphatase and alkaline phosphatase

Acid phosphatase (ACP) and alkaline phosphatase (AKP) activity was determined using the corresponding detection kits (Nanjing Jiancheng Bioengineering Institute, China) according to the manufacturer's protocols. One ACP and AKP activity unit was expressed as the production of 1 mg phenol by reaction between every 100 μ l of serum and substance in 30 min.

2.8.3. Superoxide dismutase and peroxidase

Superoxide dismutase (SOD) and peroxidase (POD) activity was assayed using the corresponding detection kits (Nanjing Jiancheng Bioengineering Institute, China) according to the manufacturer's protocols.

2.8.4. β -glucuronidase

β -glucuronidase (β -GD) was assayed using the corresponding detection kits (Nanjing Jiancheng Bioengineering Institute, China) according to the manufacturer's protocols. One β -GD unit was

expressed as the amount of phenolphthalein released per 1 h per 1 ml (U) per 1 mg of protein (U/mg). The total protein content was determined using the Bradford method, with bovine serum albumin as the standard [19].

2.9. Statistical analyses

The results in the figures are average values of 3 replicates $\pm$ standard error. The data were processed with the aid of SPSS V20.0 software and statistically analysed using analysis of variance (ANOVA). A multiple-comparison Duncan test was used to examine significant differences among all groups and a *P* value < 0.05 was considered significant.

3. Results

3.1. LC₅₀ of glyphosate on *E. sinensis*

All crabs in the control group survived during the entire experimental period. All crabs died at 48 h when exposed to 400 mg/L of glyphosate. The 24-h and 48-h LC₅₀ concentrations of glyphosate in *E. sinensis* were 461.54 and 97.89 mg/L, respectively, and the safe concentration was 4.4 mg/L.

3.2. Effect of glyphosate on THC and DHC

The mean value of THC in *E. sinensis* under normal conditions was $5.33 \pm 0.33 \times 10^6$ cells/mL, which varied but had no significant difference compared with the initial value at 0 h during the experiment. The THC decreased significantly following 6 h of exposure to each concentration of glyphosate, and the greatest decline at 42.21% was in a concentration of 98 mg/L. Then, THCs in all treatments recovered at 12 h, and there was no significant difference compared with control at 24, 48 and 96 h, respectively. In contrast, THCs in concentrations of 44 and 98 mg/L decreased continuously after 24 h and were significantly lower than that of the control at 96 h (Fig. 1).

The percentage of granulocytes in all treatments decreased and showed a significant difference compared to the control after 6 h of exposure. Afterwards, group exposure to 4.4 mg/L glyphosate recovered at 24 h, and since then, had no significant difference compared with the control, while others tended to be stable at a relatively low level. On the other hand, the percentage of

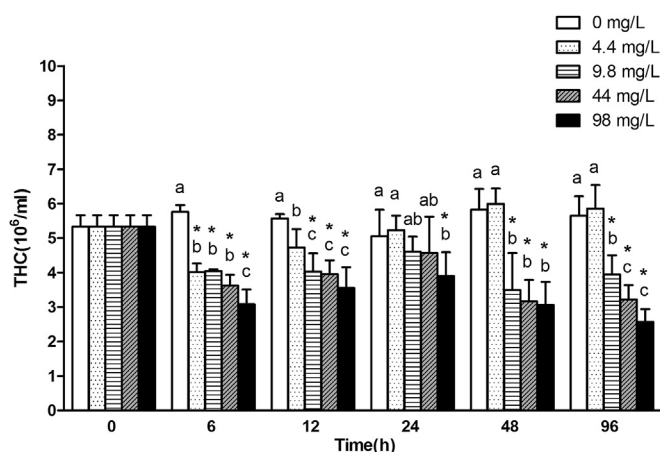


Fig. 1. Total haemocyte count (THC) from *E. sinensis* exposed to glyphosate. Note: different letters over the column represent the significant difference (<0.05) among each group, and an asterisk represents the significant difference (<0.05) between the experiment group and its initial value at 0 h.

hyalinocytes in all treatments decreased after 6 h of exposure and was significantly lower than the control at 12 h. However, indices in each group finally recovered to the normal level at 96 h (Fig. 2).

3.3. Effect of glyphosate on phagocytic activity

Phagocytosis was observed in all three types of haemocytes collected from *E. sinensis* by incubating with latex beads (Fig. 3). Hyalinocytes were the most activated phagocytic cells with a phagocytic ratio of approximately 73.21%, while semi-granulocytes and granulocytes had ratios of only 21.43% and 5.46%, respectively. After 6 h of exposure to glyphosate, phagocytic activity in all treatments decreased significantly and tended to be at the lowest level at 96 h. A phagocytic ratio of only $1.15\% \pm 0.64\%$ was determined at 96 h in a concentration of 98 mg/L (Fig. 4).

3.4. Effect of glyphosate on DNA damage of haemocytes

Several typical comet photomicrographs were observed in all treatments, especially in 44 and 98 mg/L of glyphosate at an early 24 h, and a more obvious comet tail would be seen as exposure time progresses (Fig. 5). Fig. 6 illustrated the increase in comet ratio and percentage of DNA in tails of haemocytes in each treatment from 24 h to 96 h. The comet ratio and percentage of DNA in tails exhibited a clear concentration – and time – response relationship in the exposure period, while the comet ratio had a slight decline at 96 h in concentrations of 4.4 and 9.8 mg/L. In addition, both comet ratio and percentage of DNA in tails for groups at 9.8, 44 and 98 mg/L, showed significant differences from the control at each time point (Fig. 6A and B).

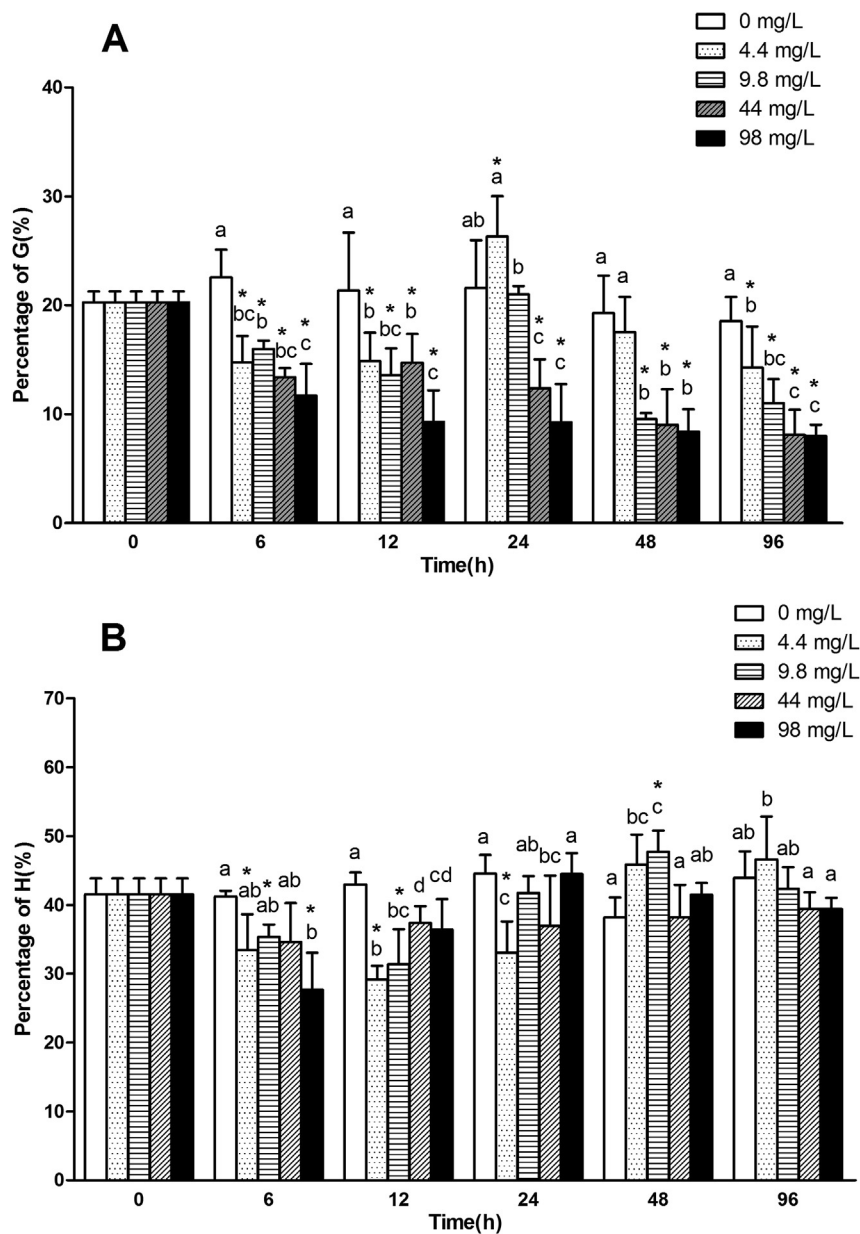


Fig. 2. Different haemocyte count (DHC) from *E. sinensis* exposed to glyphosate. A: The percentage of granulocytes (G) in haemocytes. B: The percentage of hyalinocytes (H) in haemocytes.

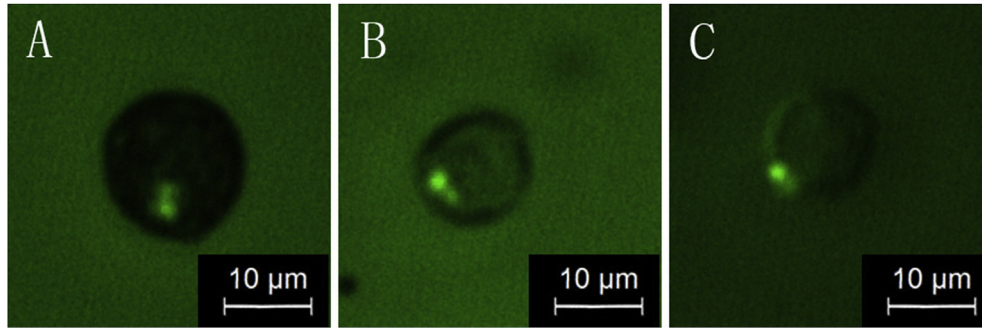


Fig. 3. The phagocytosis observed in three types of haemocytes at 400 × amplification. A: Granulocytes phagocytising latex beads. B: Semi-granulocytes phagocytising latex beads. C: Hyalinocytes phagocytising latex beads.

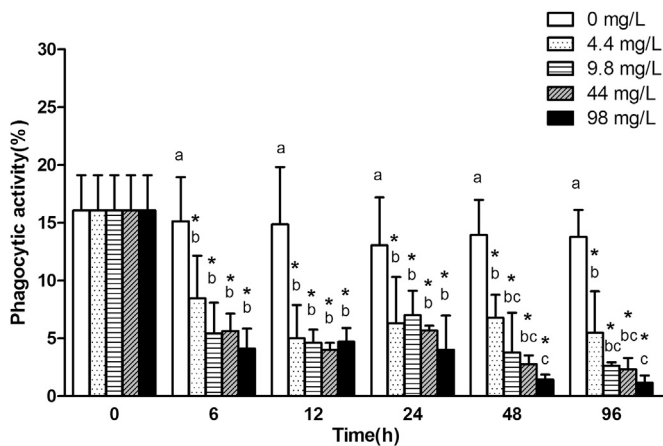


Fig. 4. Phagocytic activity of *E. sinensis* when exposed to glyphosate.

3.5. Effect of glyphosate on immune-related enzyme activity

Following exposure to glyphosate at concentrations of 98 and 44 mg/L, the ACP activity declined from 6 h to 24 h, respectively, and the ACP activity decreased gradually until the end of this experiment. ACP activity for treatments of 44 and 98 mg/L were significantly lower than others at 96 h (Fig. 7, A). Meanwhile, AKP activity in all treatments increased dramatically at 6 h but subsequently decreased. At 96 h, only the 98 mg/L-concentration group had a significant difference compared with the control (Fig. 7, B).

SOD activity in each treatment increased slightly at 6 h but markedly decreased after a temporal variation in concentrations of 4.4 and 44 mg/L at 24 h. Then, the SOD activity in the 4.4, 9.8 and 98 mg/L exposure groups restored and only the group at 98 mg/L was significant lower than the control at 96 h (Fig. 7, C). A similar

variation in the trend was observed in the POD activity of *E. sinensis*. The POD activity in the group exposed to 98 mg/L of glyphosate increased significantly at 6 h, dropped to a relatively low level at 24 h, and then tended to be stable. In a slightly different scenario, the POD activities in the 4.4 and 9.8 mg/L treatment groups increased significantly at 12 h, decreased to normal levels at 48 h, and then presented stability until 96 h (Fig. 7, D).

A remarkable increase in β -GD activities was detected in 9.8, 44 and 98 mg/L treatments, at 6 h and the β -GD activities were significantly different from that of the control at 12 h. They then tended to be stable at a high level, which was approximately double that of the control at 24 h, and all treatments at 96 h were significantly higher than both the control and their initial values at 96 h, except in a concentration of 4.4 mg/L (Fig. 7, E).

4. Discussion

Glyphosate is a phosphonoglycine herbicide that inhibits an enzyme required to produce certain amino acids, and it has a prominent toxic effect on fresh water organisms, especially fish and crustaceans [6]. However, some reports demonstrated that small crustaceans, such as amphipods, Cladocera and copepods, are not very sensitive to glyphosate, with 48 h and 96 h LC_{50} values ranging from 62 mg/L to 720 mg/L. J. S. Osterberg et al. [20] reported that the 24 h LC_{50} value of Roundup, a glyphosate-based herbicide, in juvenile blue crabs was 316 mg/L. However, the 48 h LC_{50} value of Roundup in adult *Caridina nilotica* was only 37.124 mg/L, according to P. K. Mensah's research [6]. It revealed that different crustaceans had quite different sensitivity to glyphosate, and it is therefore necessary to investigate the LC_{50} of glyphosate in important commercial species in China such as *E. sinensis*. In the present study, the 48 h LC_{50} value of glyphosate in adult *E. sinensis* was 97.89 mg/L, which is much higher than those of *Ctenopharyngodon idellus*, *Hypophthalmichthys molitrix* and *Carassius auratus* [21]. In

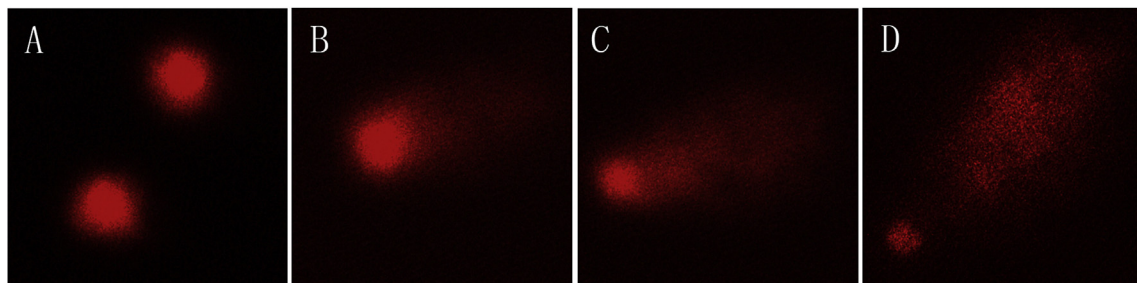


Fig. 5. Different levels of DNA damage in haemocytes from *E. sinensis* when exposed to 98 mg/L glyphosate. A: Control with no tail at 24 h; B: Comet with a tiny tail at 24 h; C: Comet with a medium tail at 48 h; D: Comet with a long tail and a smaller head at 96 h.

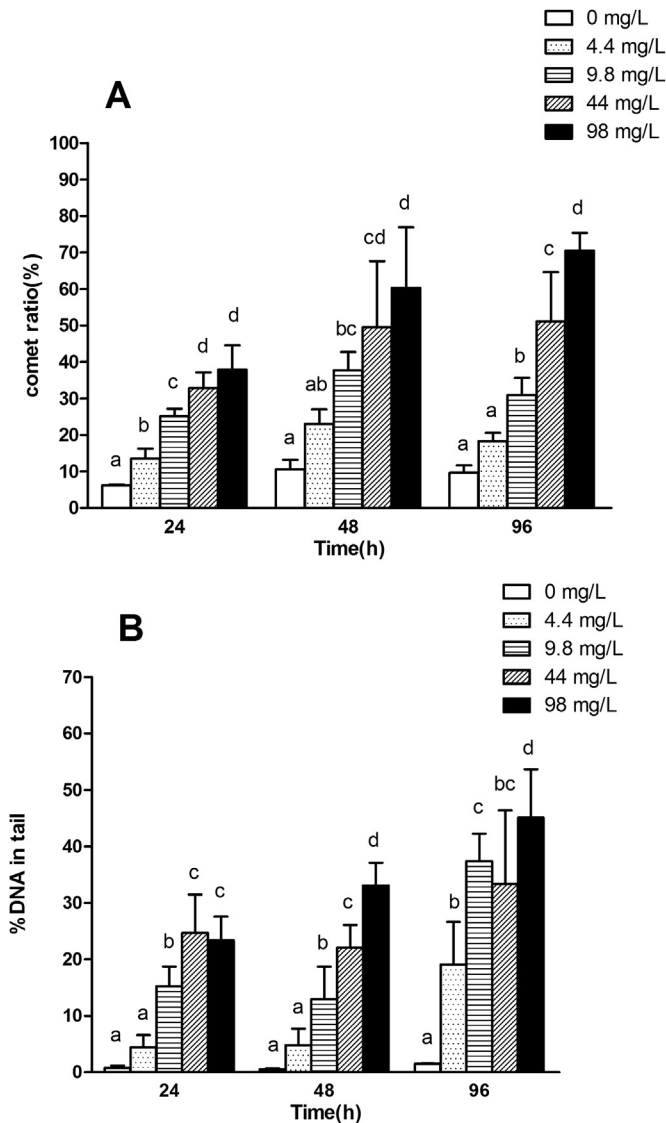


Fig. 6. DNA damage of haemocytes from *E. sinensis* when exposed to glyphosate. A: The comet ratio in haemocytes observed under a fluorescence microscope; B: The percentage of DNA in the tail calculated by CASP software.

comparison to other research, the results suggested that macrocrustacea, such as shrimps and crabs, are less sensitive to glyphosate than fishes.

As a crustacean, *E. sinensis* has a non-specific immune system involving cellular and humoral components. Haemocytes play an important role in immune response by acting as both undertakers of the cellular immunity and suppliers of the humoral immunity [11]. The THC and DHC in the haemolymph could be easily affected by extrinsic factors such as heavy metals, pathogens and pesticides [10,22,23]. In this study, the THC decreased remarkably at 6 h in all treatment groups and maintained a low level in the high concentration groups (9.8, 44 and 98 mg/L). Meanwhile, the percentages of granulocytes also decreased significantly at 6 h, and there was an apparent concentration and time response. These results were consistent with research on *Chlamys farreri* and *Squilla mantis* [11,14] following exposure to Aroclor and *Micrococcus luteus*, respectively, with the THC for both markedly decreasing at 6 h. Several studies claimed that decreases in haemocyte numbers in crustaceans might lead to impaired immune function and ultimately death [10]. A massive reduction in THC and granulocytes

was found in a sublethal concentration of glyphosate, and we considered it to possibly be the result of the lethal effects of glyphosate to *E. sinensis*.

Phagocytosis is commonly considered as one of the most important cellular immunity reactions to pathogen invasion in invertebrates. Hyalinocytes and semi-granulocytes were confirmed to be the primary phagocytic cells in haemocytes in most studies [24]. However, granulocytes and semi-granulocytes have been demonstrated to be more actively phagocytic in *Penaeus paulensis* and *Astacus astacus* [25,26]. In the current study, hyalinocytes showed strong phagocytic activity at 73.21%, while semi-granulocytes and granulocytes showed only 21.43% and 5.46%, respectively. The results were similar to a prior study on *E. sinensis* by Lv S. J. [9]. It revealed that phagocytosis in crustacean haemocytes can be affected by environmental factors, according to prior research [27]. The results in this work showed that glyphosate could significantly affect phagocytosis of haemocytes in *E. sinensis* even at a concentration of 4.4 mg/L. Phagocytic activity in each treatment decreased dramatically at 6 h and consistently decreased until 96 h. It was similar to the results of a study conducted by J. A. Xian et al. [10], which showed that phagocytic activity in haemocytes of *Penaeus monodon* decreased significantly at 6 h following exposure to 3.5 mg/L Cu^{2+} . The sharp reduction in THC and the percentage of hyalinocytes from 6 h of exposure may be responsible for the remarkable decrease in phagocytic activity during the experiment. In addition, the decrease in phagocytosis also represented a concentration and time response to glyphosate exposure in this test and suggested that THC, DHC and phagocytosis, were all sensitive parameters to evaluate the toxicity of glyphosate to *E. sinensis*.

DNA is an important germ plasm in organisms, and the damage of DNA would induce cell apoptosis or death. Thus, Comet assay is one of the most important tools for detecting DNA strand breaks to evaluate the genotoxicity of extrinsic factors. The assessment of DNA damage by comet assays after exposure to genotoxins has been reported in many aquatic animals, such as fishes and bivalves [28–30], but the only species of fresh water crustaceans studied were *Daphnia magna* and *Gammarus fossarum* [31,32] due to their widespread applications in ecotoxicological studies. In this work, the Comet assay was successfully carried out in haemocytes to evaluate the genotoxicity of glyphosate on *E. sinensis*. Apparently, DNA damage was found early at 24 h when exposed to glyphosate at a concentration of 98 mg/L with a comet ratio of $37.88 \pm 6.66\%$ and a percentage of DNA in the tail at $23.35 \pm 4.23\%$. The comet ratio and percentage of DNA in the tail were parameters usually used in Comet assays for aquatic animals, such as in the Eel *Anguilla anguilla* [33] and the Bullhead *A. nebulosus* [34]. The results of this research showed that the DNA in haemocytes were significantly damaged by exposure to even a safe concentration of glyphosate (4.4 mg/L), with a remarkable increase in the comet ratio and percentage of DNA in the tail. With the progress of exposure, both parameters increased and reached the peak at 96 h in each group. There're also dose- and time-response to glyphosate. Therefore, the comet ratio and percentage of DNA in the tail could be sensitive and accurate parameters in DNA damage detection in glyphosate exposure. In addition, the decreasing of THC and phagocytic activity had a marked time response to DNA damage from 24 h to 96 h. Consequently, the author considered that DNA damage may be the main reason for the decline in THC, which may then induce the decrease in phagocytic activity of haemocytes. The opinion was in good agreement with the research regarding DNA damage of haemocytes from *C. farreri* [11].

ACP and AKP are typical hydrolases involving extermination of extracellular invaders and are considered sensitive immune parameters in the immune response of crustaceans [15,35]. They have

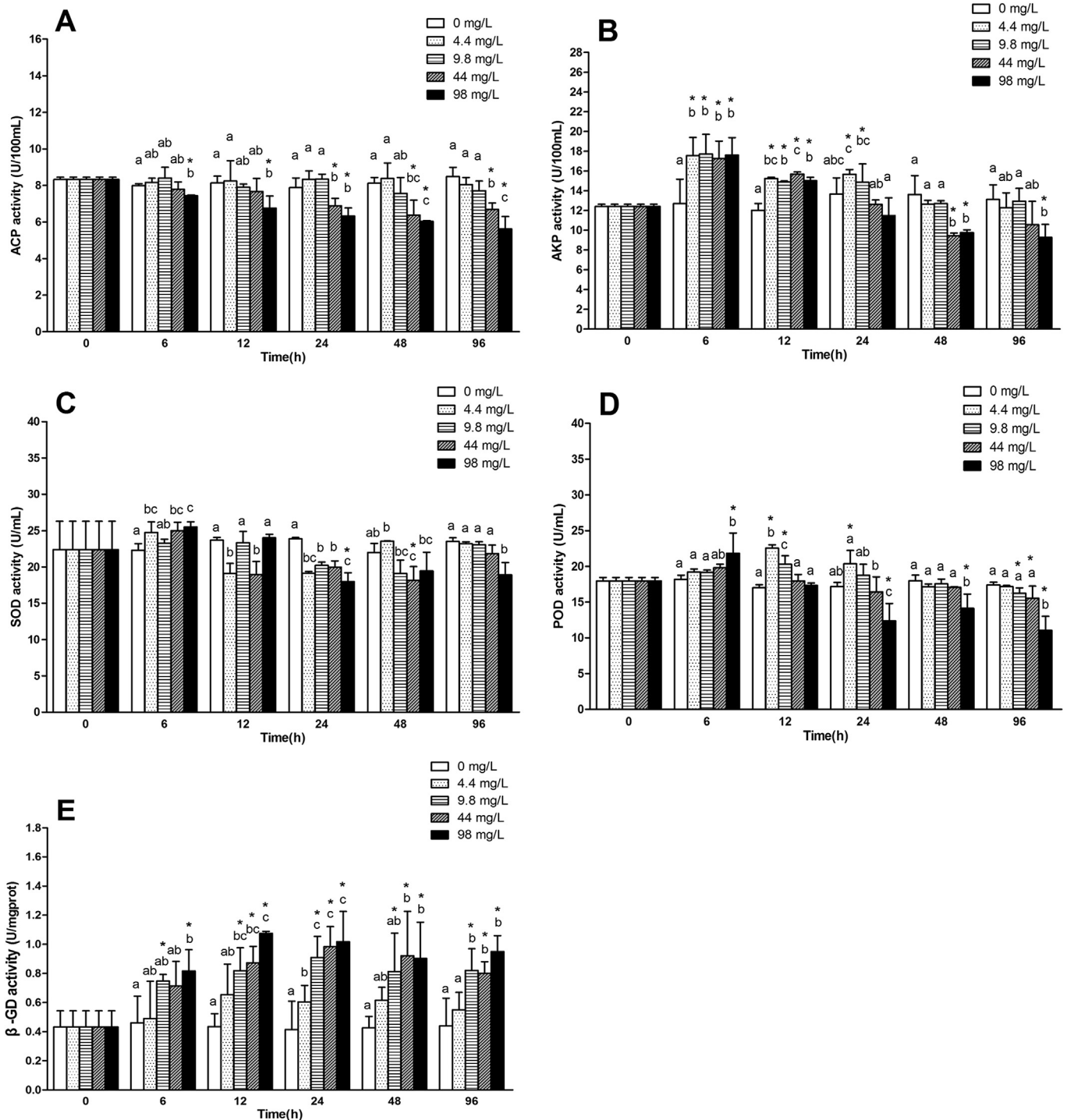


Fig. 7. Immune-related enzyme activity in serum of *E. sinensis* when exposed to glyphosate. A: ACP activity; B: AKP activity; C: SOD activity; D: POD activity; E: β -GD activity.

adaptive responses to the cytotoxic and genotoxic effects of xenobiotics and play a pivotal role in the transport of metabolites across membranes. Therefore, the enhancement of ACP and AKP levels indicated that haemocyte damage was due to disruption of membrane permeability and integrity [7]. In this study, AKP activities increased significantly in each group at 6 h but decreased in concentrations of 44 and 98 mg/L at 24 h and 48 h, respectively. The results were contrary to the observations by S. Dey that the AKP activity of three freshwater teleosts increased after glyphosate

exposure at a concentration of 1.2 mg/L. However, a reduction in activity of AKP was also reported by Arellano et al. [36] in the sub-lethal Copper exposure in *Solea senegalensis*. Jiraungkoorskul et al. [37] suggested that the variation in AKP activity was a result of physiological and functional alterations in fish exposure to herbicides. Therefore, we suggested that low concentrations of glyphosate may induce haemocyte apoptosis by DNA damage to release ACP and AKP in the serum, which is recovered with immune response. However, in high concentrations, for example, the sub-

lethal concentration would induce an inhibition of the mechanism of enzymatic synthesis. More work should be conducted in later research.

It is reported that pesticides may cause oxidative stress by the generation of some reactive molecules, including reactive oxygen species (ROS) [38]. Too much ROS could induce a series of tissue damage, and the antioxidation system is necessary for clearing excess ROS. SOD and POD are important enzymes in antioxidation for the clearance of O_2^- and H_2O_2 and became common parameters in oxidative damage assessment. In the present work, increases in SOD and POD were found in 98 mg/L at 6 h, and the values then decreased from 24 h while no significant change occurred in 4.4 and 9.8 mg/L. The results were similar to the investigation of *Macrobrachium rosenbergii* [16], which showed that SOD activity decreased after a dopamine challenge at 2 h. It considered that high concentrations of glyphosate may inhibit SOD and POD activity causing serious oxidation damage.

β -GD is known to be the mediator of colon cancer, and β -GD activity was usually measured as a parameter to evaluate the risk of colorectal cancer in humans [39]. On the other hand, β -GD is also regarded as a biomarker candidate for the assessment of low-level organophosphorus pesticide exposure to mammals such as rats and humans. Research from J. Ueyama et al. [40] demonstrated that β -GD detection in plasma was a practical approach and a more sensitive parameter for the biomonitoring of organophosphorus pesticide exposure to humans compared with plasma butyrylcholinesterase (BChE). C. Gallo et al. [14] located β -GD in granules in three types of haemocytes in *Squilla mantis* by a cytoenzymatic stain. However, the β -GD activity was detected in this test for the purpose of assessing toxicity of glyphosate to the freshwater crustacean *E. sinensis* for the first time. The results showed that the β -GD activity in all treatments had an increase at 6 h and tended to be stable at 24 h. All treatment groups, except that of 4.4 mg/L, were significantly higher than the control at 96 h. The results were consistent with studies from J. Ueyama et al. [40], which showed that high levels of β -GD indicate more exposure to organophosphorus pesticides in humans. Furthermore, we found that a higher exposure concentration lead to higher β -GD activity and there was a marked dose response between glyphosate exposure and β -GD activity in *E. sinensis*. There was also a time response within 12 h of exposure. Therefore, the author suggested that β -GD activity could be an effective biomarker to evaluate the toxicity of glyphosate to *E. sinensis* or other freshwater crustaceans.

5. Conclusion

In this study, glyphosate had a significant effect on immune response, including THC, DHC, phagocytosis, immune-related enzyme activity and haemocyte DNA damage in *E. sinensis*. Even at the safe concentration of 4.4 mg/L, glyphosate could induce slight DNA damage and immune inhibition, especially in haemocyte counts, phagocytosis, AKP activity and SOD activity within 6 h. It would increase the risk of pathogen invasion as the host is under a sub-health status. Therefore, it suggested that it is better to not use glyphosate in aquatic water. In addition, THC, haemocyte phagocytosis activity, haemocyte DNA damage and β -glucuronidase activity, were considered more sensitive biomarkers for toxicity assessment of glyphosate exposure to *E. sinensis*, as well as possibly other freshwater crustaceans.

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